

Event vs full-path vs cycle replanning for three planners

Rebuilding more often costs more and finishes fewer runs: at 20 ms checks Visibility Graph rebuilds 23.2 (event), 46.2 (full-path) and 107.6 (cycle) times per run. Across all planners, failed runs from a rebuild over 100 ms (computing plus waiting for the CPU, as a background process): event 30, full-path 81, cycle 70; failed runs because no route existed for 1 s (obstacles blocked every path): 2, 10, 7. Event (ours) rebuilds when the next segment is blocked; full-path also when any later segment is (Costa and Tonidandel 2024); cycle rebuilds at every check. Brackets show the change from event. In each group of three, ▲ blue (bold) marks the best value and ▼ orange the worst; planning time and time per rebuild are marked only when they differ by more than 20 %, the drift between batches. The rebuild-over-100 ms column depends on the machine; the no-route column does not.

	Runs finished (of 200)	Finished without contact (of 200)	Rebuilds per run	Rebuilds from blocked later segment (/run)	Planning time per run (ms)	Time per rebuild, median (ms)	Failed runs: rebuild over 100 ms	Failed runs: no route for 1 s	Time a route is kept (ms)
Voronoi									
Event (ours) @ 20 ms	198 ▲	140	9.1 ▲	0.7	107 ▲	5.7	2 ▲	0	518 ▲
Full-path @ 20 ms	197 (−1)	137 (−3) ▼	9.7 (+0.6)	0.6 (−0.2)	158 (+51)	6.7 (+1.0)	3 (+1)	0	513 (−5)
Cycle @ 20 ms	196 (−2) ▼	145 (+5) ▲	110.4 (+101.3) ▼	–	727 (+620) ▼	5.6 (−0.1)	4 (+2) ▼	0	110 (−409) ▼
Event (ours) @ 100 ms	199 ▲	91 ▼	5.2	0.3	44 ▲	5.8	1 ▲	0	789
Full-path @ 100 ms	198 (−1)	91 ▼	5.2 ▲	0.3	64 (+20)	6.7 (+0.9)	2 (+1)	0	791 (+3) ▲
Cycle @ 100 ms	195 (−4) ▼	95 (+4) ▲	22.0 (+16.8) ▼	–	155 (+111) ▼	5.8	5 (+4) ▼	0	405 (−384) ▼
PRM									
Event (ours) @ 20 ms	194 ▲	142	7.7 ▲	0.0 ▲	120 ▲	10.6	6 ▲	0 ▲	560 ▲
Full-path @ 20 ms	181 (−13) ▼	135 (−7) ▼	12.5 (+4.8)	4.5 (+4.5) ▼	262 (+142)	12.6 (+1.9)	15 (+9) ▼	4 (+4) ▼	433 (−127)
Cycle @ 20 ms	182 (−12)	144 (+2) ▲	75.9 (+68.2) ▼	–	1084 (+965) ▼	10.6 (−0.1)	15 (+9) ▼	3 (+3)	145 (−416) ▼
Event (ours) @ 100 ms	194 ▲	90 ▼	4.6 ▲	0.0 ▲	63 ▲	10.6	5 ▲	1 ▲	781 ▲
Full-path @ 100 ms	191 (−3) ▼	92 (+2)	7.1 (+2.5)	2.8 (+2.8) ▼	141 (+78)	12.8 (+2.2) ▼	7 (+2) ▼	2 (+1)	678 (−103)
Cycle @ 100 ms	191 (−3) ▼	93 (+3) ▲	14.9 (+10.3) ▼	–	216 (+153) ▼	10.5 (−0.1) ▲	6 (+1)	3 (+2) ▼	486 (−296) ▼
Visibility Graph									
Event (ours) @ 20 ms	187 ▲	133 ▲	23.2 ▲	0.0 ▲	424 ▲	14.2	12 ▲	1 ▲	341 ▲
Full-path @ 20 ms	162 (−25) ▼	114 (−19) ▼	46.2 (+22.9)	25.5 (+25.5) ▼	998 (+574)	16.5 (+2.3) ▼	36 (+24) ▼	2 (+1) ▼	205 (−136)
Cycle @ 20 ms	169 (−18)	119 (−14)	107.6 (+84.3) ▼	–	1667 (+1243) ▼	13.6 (−0.6) ▲	30 (+18)	1 ▲	106 (−235) ▼
Event (ours) @ 100 ms	196 ▲	82 ▲	8.3 ▲	0.0 ▲	141 ▲	13.7	4 ▲	0 ▲	658 ▲
Full-path @ 100 ms	180 (−16) ▼	75 (−7) ▼	14.0 (+5.7)	6.0 (+6.0) ▼	308 (+168)	16.3 (+2.6)	18 (+14) ▼	2 (+2) ▼	509 (−149)
Cycle @ 100 ms	190 (−6)	80 (−2)	21.0 (+12.7) ▼	–	353 (+212) ▼	13.9 (+0.2)	10 (+6)	0 ▲	412 (−246) ▼